Acknowledgment

I take this opportunity to express my heartiest gratitude to many who extended their helping hand and inspiration during the project. The successful completion of this project work stands on the shoulder of the people who have helped us directly or indirectly.

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I am indebted to my other faculty members and an internal guide for motivating me

.

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Finally, I would like to thank everyone who directly or indirectly helped us in the project.

With Thanks to All.

Kashish patni

Krisa shah

Dhruvi Panchal

Vidhi Prajapati

Khushi shah

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1. <u>Project profile</u>

Title	description
Project title	Fun zone games
IDE	Visual studio code
Team size	5 members
Team members	Kashish patni(31) Krisa shah(46) Dhruvi Panchal(22) Vidhi Prajapati(35) Khushi shah(45)
Guide	Dr.Bhumika shah
Project duration	6 months

3.Project description

FUN ZONE GAMES

1. Flappy Bird
2. Memory Game
3. Block Puzzle

1. Flappy Bird

How to play Game ?

- Press the start button to start the game
- Press the space bar key ,upper key ,and X key to fly the bird
- The bird automatically falls down
- Control the bird through the pipes with random heights
- Fly as far as possible to get high scores
- The game will over when the bird hits one of the pipes ,ground, or ceiling
- Press restart button to play game again

2. Memory Game

- Classic card-matching game .
- Players need to find pairs of matching cards .
- Three levels of this game easy , medium , hard.
- Made for those who enjoy memory and puzzle challenges.
- Player can chose the level :easy , medium , hard.
- Player will click on two card one by one.
- If two card are matched they will be kept open otherwise both card close.
- when player completes the level .the time and moves will be displayed on screen .
- When two cards will be open then show moves 1 .
- The winning message will be displayed .
- There is a option to go to next level or go to main page.

3. Block game

- The classic grid-based puzzle game.
- Set blocks on the board .
- You can drag & drop blocks onto grid .
- blocks are placed in lines , rows, and grids completely.
- When you complete a row you will clear the blocks from the board.
- If blocks will not disappear quickly the board will start to fill.
- When the block reach the top of the field and there is no space for new blocks to arrive the game ends there.

4. Related work (literature review)

1. **Flappy Bird**

- **summary**

Flappy Bird is an interesting game where players navigate a bird through a series of pipes by tapping the screen to make it flap. This game is known for its simplest yet challenging gameplay, the objective is to achieve the highest score by avoiding obstacles.

- **Limitation**

1.1 Single Player :-

This game allows a single player gameplay. Thus it collapse the game as the per the demand of multiplayer.

- **Uniqueness**

1.1 Challenging Gameplay:

Navigating through narrow gaps created a difficult but good experience.

2.Memory Game

- **Summary**

The Memory Game is a challenging game where players flip cards to find matching pairs. By testing memory recall, it exercises the brain and enhances concentration. The objective is to find all pairs in the least attempts. With increasing difficulty, it provides a interesting and enjoyable experience for players of all ages.

- Memory game focus on short-term memory, which removes other crucial aspects of cognitive function and real-world memory challenges.
- It Enhances recall, sharpens focus, and challenges cognitive abilities uniquely.

2.Block game

- Summary:

Block puzzle games challenge players to fit various shapes together within a given space. The goal is to create complete rows or columns, causing them to disappear. As levels progress, the puzzles become more complex, requiring strategic thinking and awareness. It's a satisfying blend of logic and creativity in a digital puzzle format.

- Uniqueness:

"Block puzzle game stands out with fun challenges, strategic play, and endless creative possibilities."

5. Tools and technologies

IDE	Visual studio code
Graphics software	Adobe photoshop
Version control	Github
Web browser	Chrome

Description of the tools and technologies:

1.IDE

Used integrated development environment(IDE) visual studio code for writing html,css,javascript

2.graphics software

Used adobe photoshop for editing game assests and background

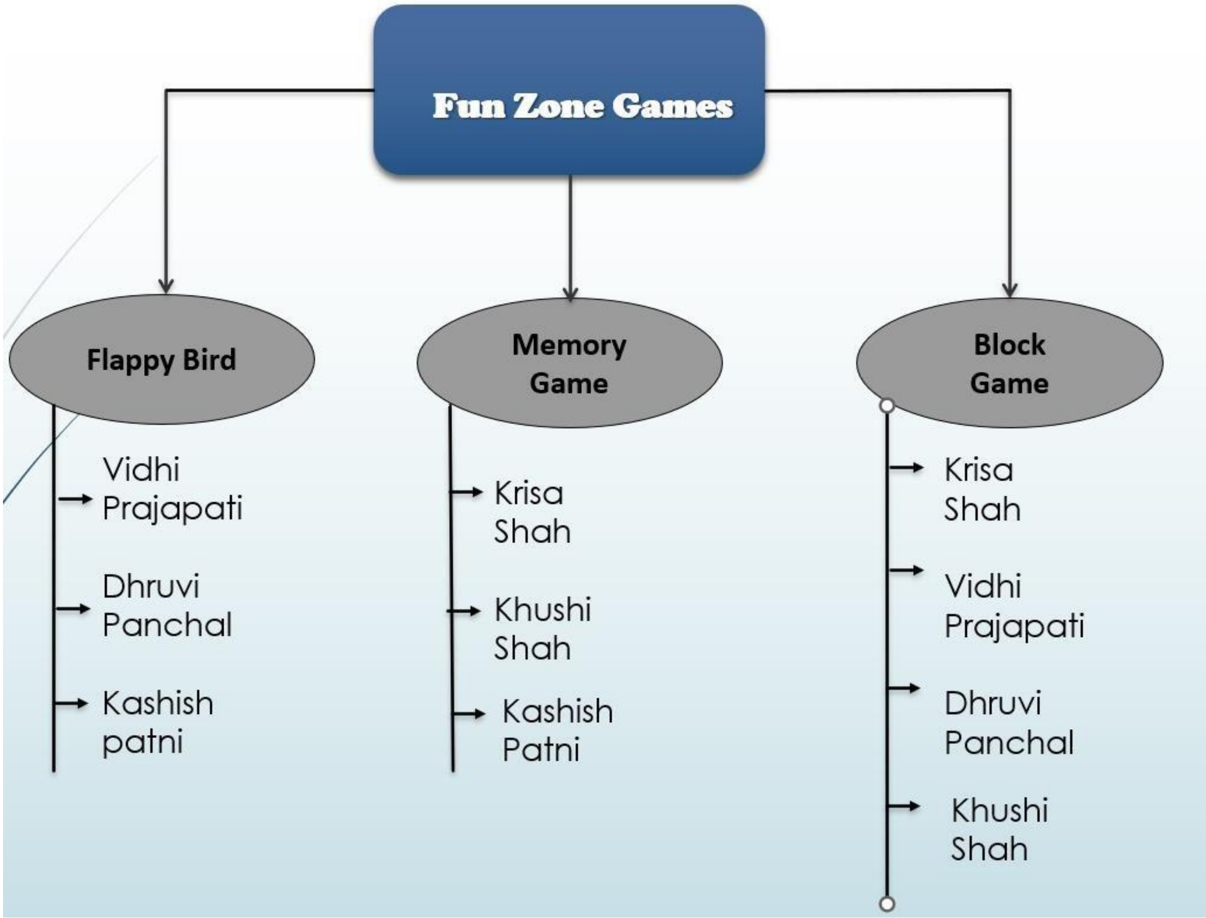
3.version control

Used github for version control and collaboration

4.web browser

Used for testing and debugging the game in chrome.

6. Work division



7. Proposed system

HTML Structure:

Designed a clear and organized HTML structure for each game, including elements for the game canvas, score display, and any other necessary components.

CSS Styling:

Applied CSS styling to create an appealing and responsive layout for the games.

JavaScript Logic:

Implemented the game mechanics and logic using JavaScript. This includes handling user input, updating game state, and graphics on the canvas.

Game Loop:

Sated up a game loop to ensure continuous updates , providing a smooth and interactive experience for players.

Collision Detection:

Created a functions for detecting collisions between game objects, essential for gameplay in both Flappy Bird and the block puzzle.

Scoring System:

Developed a scoring system for each game. This could involve tracking score, levels, or progress based on user performance.

Version Control:

Use version control systems like Git to track changes and collaborate with others efficiently.

8 . Screenshot / animation

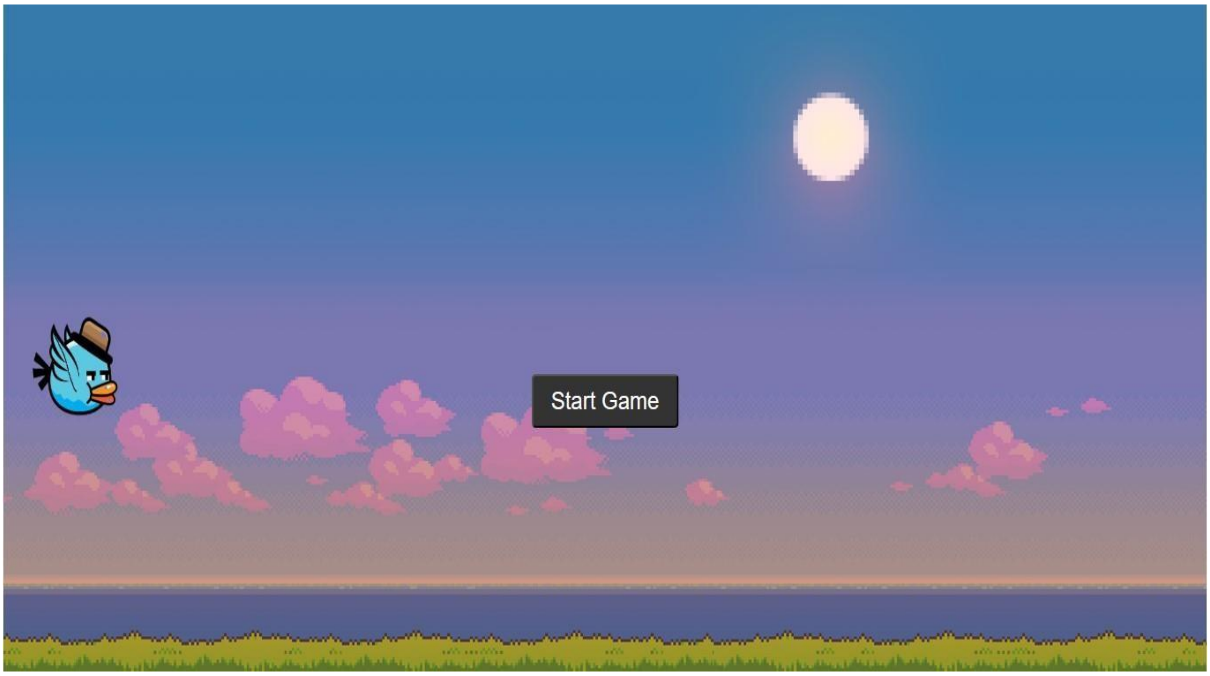
8.1 Title page

8.1 This is the title page of out fun zone games.

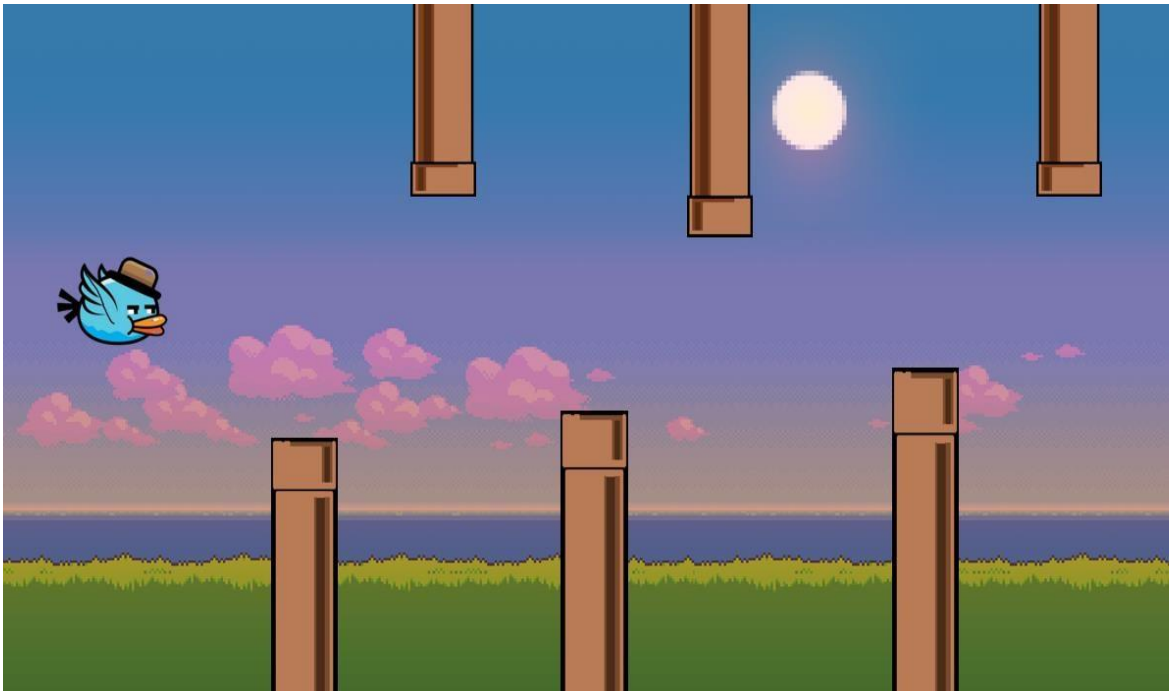
8.2 Main page

8.2 This is the main page of our game where three games are displaying

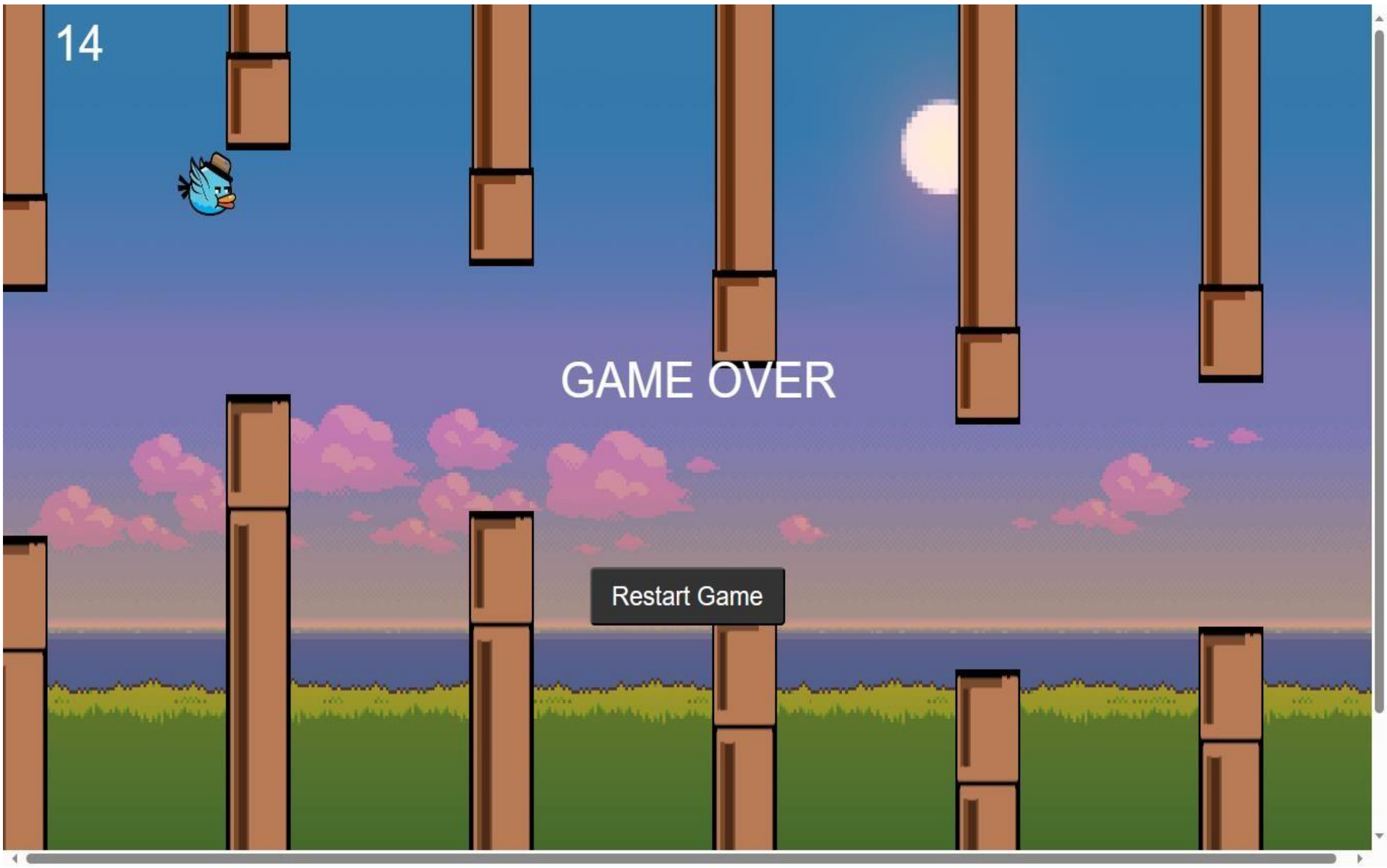
8.3 Flappy bird



8.3.1 This is a screenshot of out flappy bird game



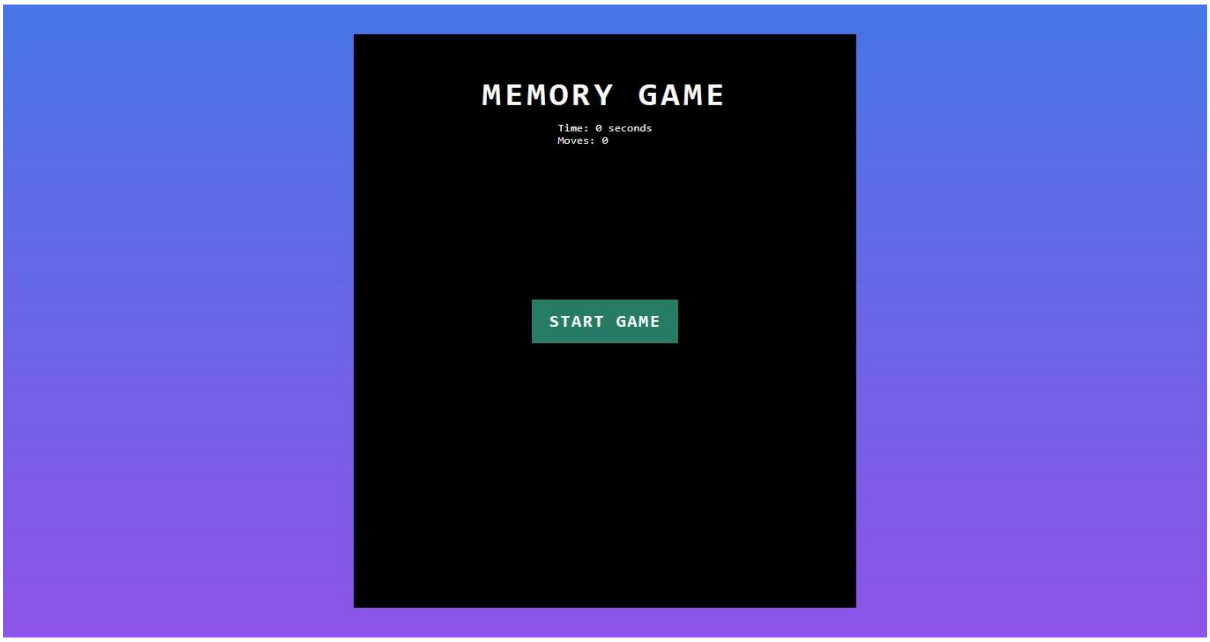
8.3.2 This is the screenshot of flappy bird game when game starts



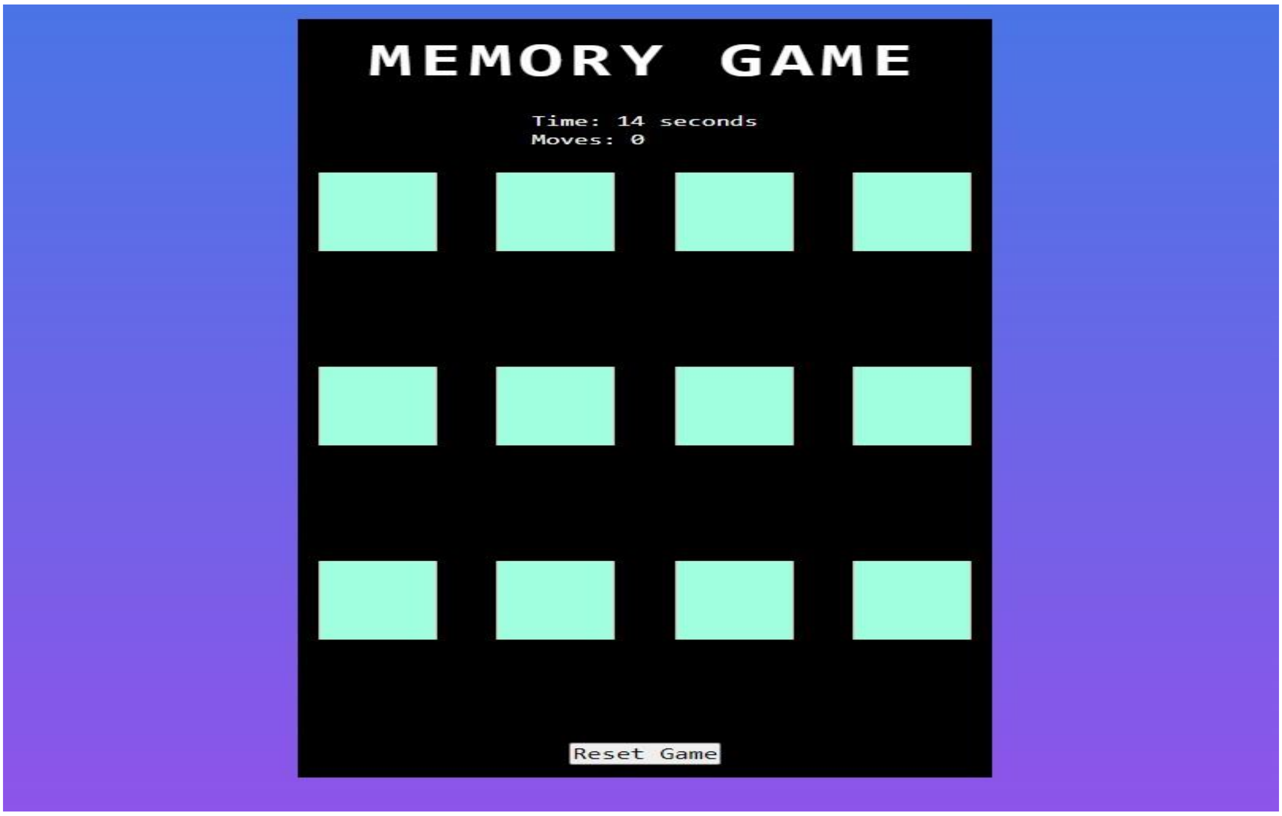
8.3.3 This is the screenshot of out flappy bird game when game is over

8.4 Memory game

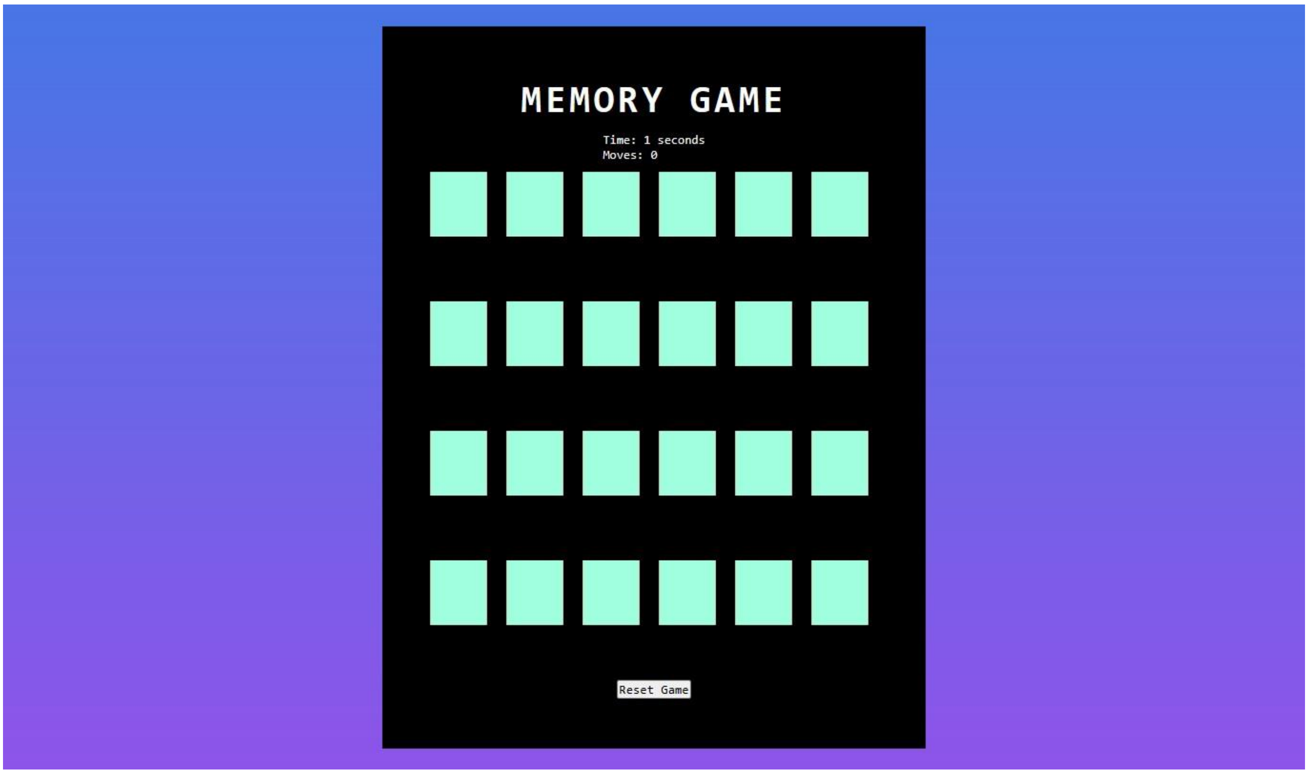
8.4.1 This is the screenshot of our memory game



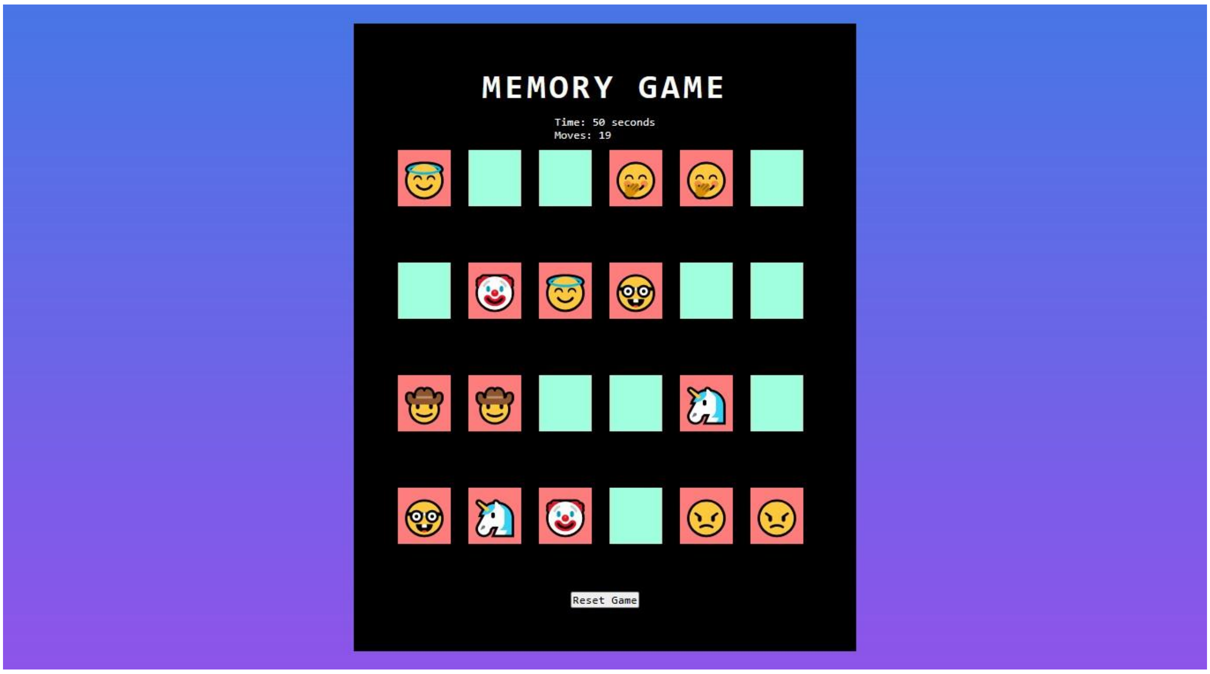
8.4.2 This is the screenshot of memory game where you can start the game by clicking on the button



8.4.3 This is the screenshot of memory game(easy level)



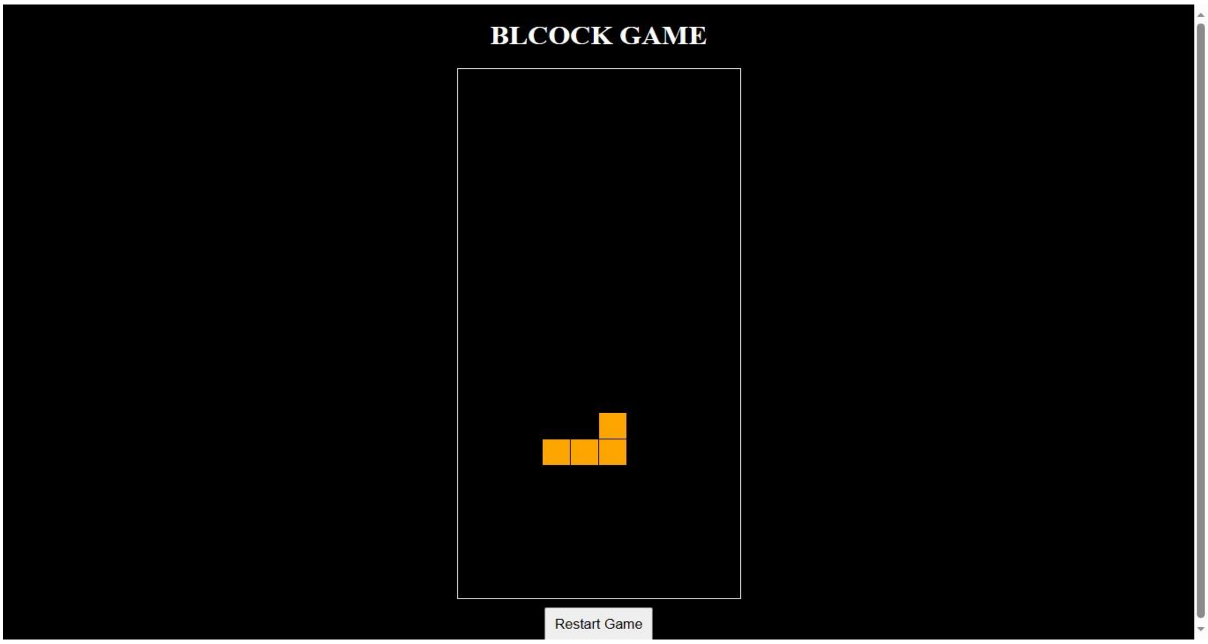
8.4.4 This is the screenshot of memory game (hard level)



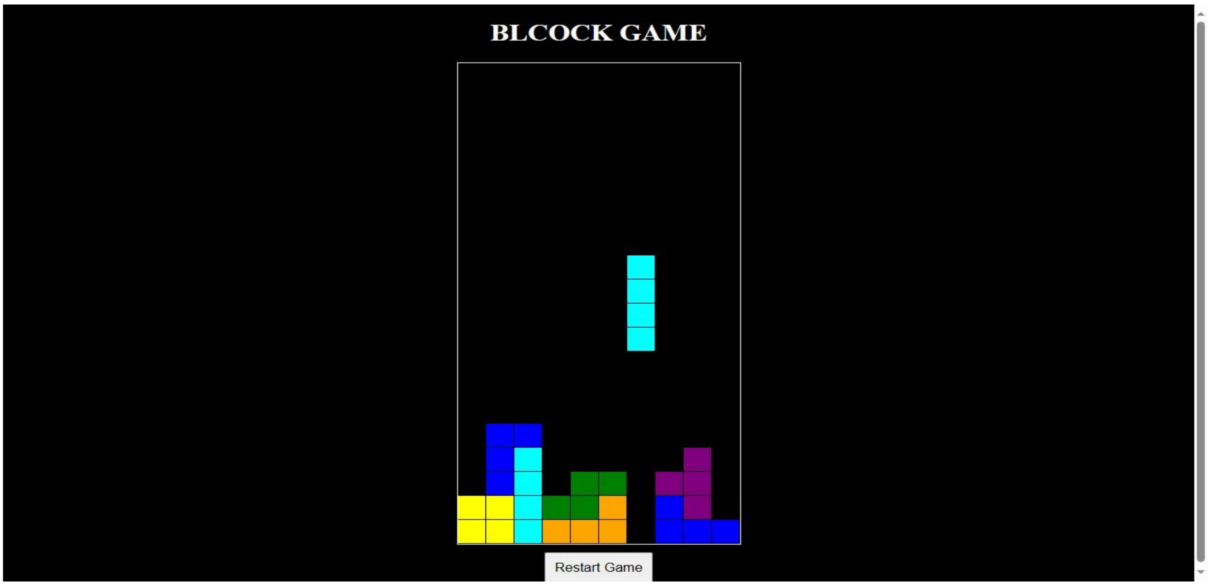
8.4.5 This is the screenshot of memory game where half cards are open and half close

8.4.6 This is the screenshot of memory game where game is completed

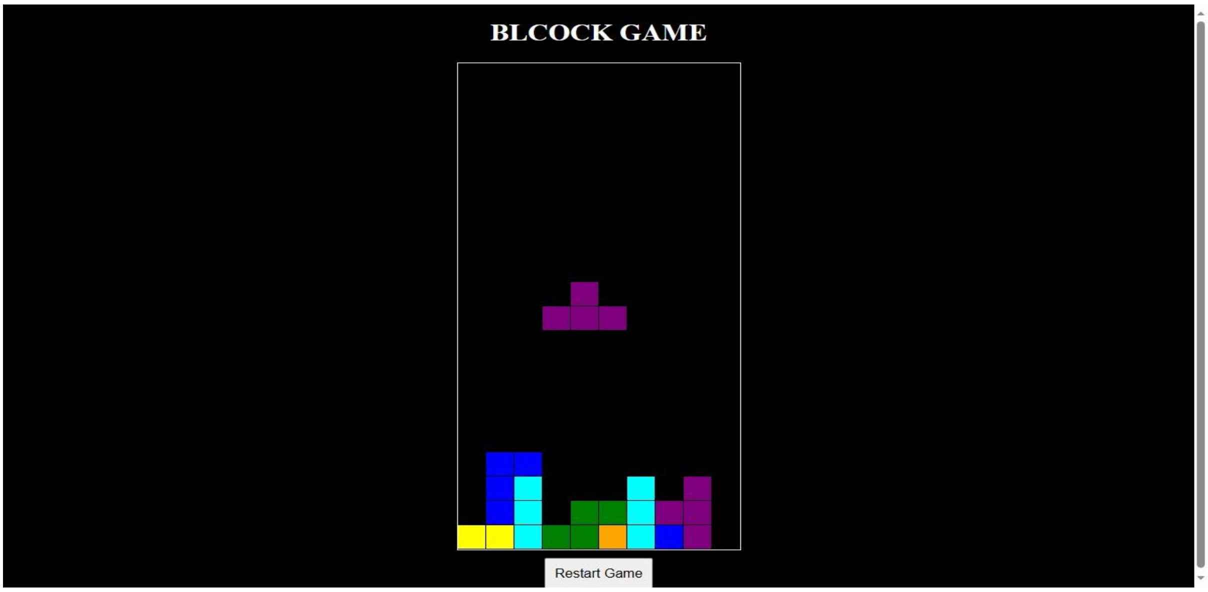
8.5 Block game



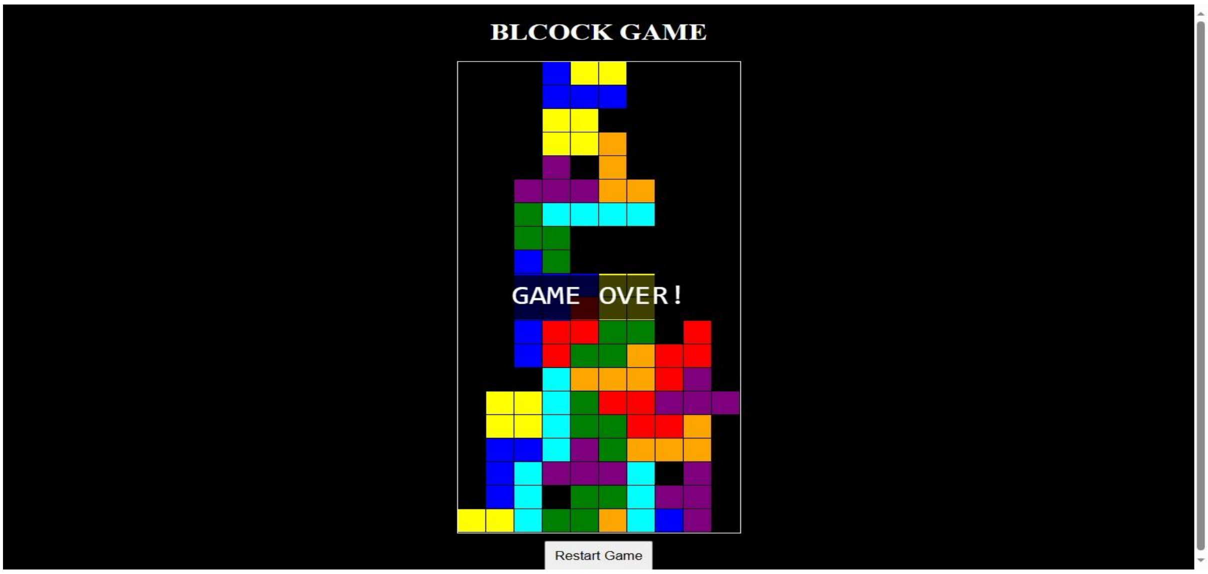
8.5.1 This is the screenshot of block game



8.5.2 This is the screenshot of block game where game is going on `



8.5.3 This is the screenshot of block game where game is goin on



8.5.4 This is the screenshot of memory game where game is over.

9. Implementation

9.1 Code snippet of memory game

```
function checkMatch(card1, card2) {
    if (card1.innerHTML === card2.innerHTML) {
        card1.classList.add("boxMatch");
        card2.classList.add("boxMatch");
        openedCards = [];
        if (document.querySelectorAll('.boxMatch').length === emojis.length) {
            clearInterval(timerInterval);
            winMessage.style.display = "block";
        }
    } else {
        setTimeout(() => {
            card1.classList.remove('boxOpen');
            card2.classList.remove('boxOpen');
            openedCards = [];
        }, 500);
    }
}
```

- The function takes two parameters, card1 and card2 .It is representing two cards that are being compared for a match.
- It compares the innerHTML property of card1 and card2. If the contents are same, it is considered a match.
- If the cards match, add the "boxMatch" class to both cards.Reset the openedCards array to an empty array.Check if the number of elements with the class 'boxMatch' is equal to the total number of emojis .If so, then stop the game timer and display a win message.
- If the cards do not match, after a delay of 500 milliseconds, remove the 'boxOpen' class from both cards.Reset the openedCards array.
- This function seems to be an essential part of the logic for checking card matches in a memory or matching , where players flip cards to find matching pairs.

9.2 Code snippet of flappy bird game

```
function moveBird(e) {
    if (e.code == "Space" || e.code == "ArrowUp" || e.code == "KeyX") {
        //jump
        velocityY = -6;


        //reset game
        if (gameOver) {
            bird.y = birdY;
            pipeArray = [];
            score = 0;
            gameOver = false;


        }
    }
}
```

Explanation:

- The function movebird takes an event parameter e.
- It checks if the key code of the pressed key is either "Space", "ArrowUp", or "KeyX".
- If one of the specified keys is pressed, it sets the velocityY to -6. This represents a jump or upward movement for the bird.
- If the game is over , it performs the following actions:

  1. Resets the bird's vertical position to the initial position (birdY).

  2.Clears the pipeArray, which may contain information about obstacles or pipes in the game.

  3.Resets the score to 0.

  4.Sets gameOver to false, indicating that the game is not over.

This function seems to handle both the bird's jumping action and the game reset logic when specific keys are pressed.

9.3 Code snippet of block puzzle game

```
// game loop
function loop()
{
  rAF = requestAnimationFrame(loop);

  context.clearRect(0, 0, canvas.width, canvas.height);

  // draw the playfield
  for (let row = 0; row < 20; row++)
  {
    for (let col = 0; col < 10; col++)
    {
      if (playfield[row][col])
      {
        const name = playfield[row][col];
        context.fillStyle = colors[name];

        // drawing 1 px smaller than the grid creates a grid effect
        context.fillRect(col * grid, row * grid, grid - 1, grid - 1);
      }
    }
  }
```

- Function loop starts and `requestAnimationFrame` function to schedule the next iteration of the game loop.
- context.clearRect This line clears the entire canvas before redrawing the game state in the next frame.
- It ensures that the previous frame's content is removed, preventing the accumulation of drawings.
- Drawing the Playfield:
- The loop iterates over the rows of the playfield.
- The nested loop iterates over the columns of the playfield.
- if conditon checks if there is a block (non-empty) at the specified row and column in the playfield.
- Then Retrieves the name of the block or element at the current position in the playfield.
- context.fillStyle Sets the fill colour of the drawing context to the color associated with the block type .
- Draws a filled rectangle representing the block on the canvas. The position and dimensions are determined by the current row and column, and the size of each grid cell (`grid`). Subtracting 1 from the width and height creates a grid effect by leaving a small gap between adjacent rectangles.

10. Learning outcomes

- gained the understanding of HTML,CSS,JAVASCRIPT
- learnt the CANVAS to draw graphics, animation and interactive gaming environment
- learnt the logic building for the game loop, collison detection.
- Learnt the version control system like git for tracking changes, collaborating with others and maintaining a projects history
- By working in a team,we improved collaboration and communication skills by coordinating task,providing updates.

11. Future work and conclusion

- This project highlights the creative and technical aspects of game development, show the ability to create enjoyable and interactive games for diverse audiences**.**
- For future work we can add different types of sounds for the different actions .we can add more levels In the game to increase difficulty

12. Bibliography

https://github.com/vidhiprajapatii/project_funzone.gzt -

Reference:

1.W3school

2.Git hub